The BBDG Decomposition Thm:

§0: Smooth case:

Deligne proved:

Thm. $f: X \to Y$ smooth, projective. Then $Rf_* \underline{\mathbb{Q}}_X \cong \bigoplus_{i \geq 0} \mathcal{H}^i(Rf_* \underline{\mathbb{Q}}_X)[-i]$

and $\mathcal{H}^i(Rf_* \underline{\mathbb{Q}}_X) = R^i f_* \underline{\mathbb{Q}}_X$ is a semisimple local system $\forall i$.

Pf Sketch:

(Hard Lefschetz): Suppose $K \in D^b_c(Y)$ has a map $L: K \to K[2]$. s.t.

$L^i : \mathcal{H}^{-i}(K) \xrightarrow{\sim} \mathcal{H}^i(K)$, $\mathcal{H}^i(K)$ in degs $[-n, n]$.

$A = \mathcal{H}^{-n}(K)$, $B = \mathcal{H}^n(K)$.

$$\Rightarrow \quad A[n] \xrightarrow{a} K \xrightarrow{L^n} K[2n] \xrightarrow{b} B[n]$$

$\sim$ from H.L.

$\Rightarrow$ a has retraction $K \xrightarrow{b \circ L^n} B[n] \xrightarrow{(HL)^{-1}} A[n]$

b has section $B[n] \xrightarrow{(HL)^{-1}} A[n] \xrightarrow{L^n \circ a} K[2n]$.

$\Rightarrow$ $\mathcal{H}^{-n}(K)$ & $\mathcal{H}^n(K)$ split off & can continue from there.

$\Rightarrow$ $Rf_* \underline{\mathbb{Q}}_X \cong \bigoplus_{i \geq 0} R^i f_* \underline{\mathbb{Q}}_X[-i]$.

Semisimplicity of $R^i f_* \underline{\mathbb{Q}}_X$ can be proved using either:

- ▸ Hodge theory / Riemann-Hilbert: relate to D-modules & use Hodge theory of D-modules
- ▸ Positive characteristic: use Weil conjectures + spreading out.

We will return to these later after some applications...

Same pf. shows that if $f: X \to Y$ proper & X is smooth; then $Rf_* \underline{\mathbb{Q}}_X \cong \bigoplus_{i \geq 0} {}^p\mathcal{H}^i(Rf_* \underline{\mathbb{Q}}_X)[-i]$.

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Applications of Deligne's Thm:

Cor. $f: X \to Y$ smooth, proj. Then $H^*(X,\mathbb{Q}) \cong \bigoplus_{i\geq 0} H^{*-i}(Y, R^i f_* \underline{\mathbb{Q}}_X)$

and the Leray spectral seq:

$$E_2^{p,q} = H^p(Y, R^q f_* \underline{\mathbb{Q}}_X) \Longrightarrow H^{p+q}(X,\mathbb{Q})$$

degenerates on page 2.

Pf: Follows bc $\tau_{\leq q-1} K \to \tau_{\leq q} K \to \mathcal{H}^q(K)[-q] \xrightarrow{[1]}$ rs split $\forall q$. //

Cor: $H^q(X,\mathbb{Q}) \twoheadrightarrow H^0(Y, R^q f_* \underline{\mathbb{Q}}_X) \cong H^q(F,\mathbb{Q})^{\pi_1(Y)} \subseteq H^q(F,\mathbb{Q})$

where $F =$ (fiber of f)

In fact we will later show more!

Global Invariant Cycles Thm: $X \subseteq_{\text{cpt}} \overline{X}$. Then

$$\text{Image}\left(H^*(\overline{X},\mathbb{Q}) \to H^*(F,\mathbb{Q})\right) = \text{Image}\left(H^*(X,\mathbb{Q}) \to H^*(F,\mathbb{Q})\right)$$
$$= H^*(F,\mathbb{Q})^{\pi_1(Y)}$$

Hodge str. $\rightsquigarrow$ Hodge str.

Cor (Projective Bundle Formula). $E \to X$: rk r vector bdl. Then

$H^k(\mathbb{P}(E),\mathbb{Q}) \underset{(\star)}{\cong} \bigoplus_{j=0}^{r-1} H^{k-2j}(X,\mathbb{Q})$. (not as a ring!)

Pf: $f^*\mathbb{P}(E) \twoheadrightarrow \mathcal{O}_{\mathbb{P}(E)}(1)$ $\rightsquigarrow \xi = c_1(\mathcal{O}_{\mathbb{P}(E)}(1))$ gives hyperplane class on each fiber

$f: \mathbb{P}(E) \to X$

$\Rightarrow \xi^j =$ global gen. for $R^{2j} f_* \mathbb{Q}$

Vijer: $R^{2j} f_* \mathbb{Q} = \underline{\mathbb{Q}}_X$, $R^{2j+1} f_* \mathbb{Q} = 0$ $\Longrightarrow Rf_* \mathbb{Q}_{\mathbb{P}(E)} \underset{\text{Deligne}}{\cong} \bigoplus_{j=0}^{r-1} \underline{\mathbb{Q}}_X[-2j]$

$\overset{\text{hyper}}{\underset{\text{-coh}}{\Longrightarrow}}$ $(\star)$ //

Some first examples of Decomp^n Thm:

Prop. $f: X \to Y$ finite map between curves. Then $Rf_*\underline{\mathbb{Q}}_X[1]$ is equal to $j_{!*}(\mathcal{L}[1])$ for $\mathcal{L} = R^0 f_* \underline{\mathbb{Q}}_X|_U$ & $U \xrightarrow{j} X$ is opens small enough $\mathcal{L} \in \mathrm{Loc}(U)$.

Pf: $U := Y - (\text{branch pts})$

$j^*(Rf_*\underline{\mathbb{Q}}_X[1]) = \mathcal{L}[1]$ is clear.

$i_y : \{y\} \hookrightarrow Y$ for $y \in Y - U$.

$i_y^*(Rf_*\underline{\mathbb{Q}}_X[1]) \overset{\text{proper base change}}{\cong} R\Gamma(f^{-1}(y), k)[1] \cong k^{\#f^{-1}(y)}[1]$

$i_y^!(Rf_*\underline{\mathbb{Q}}_X[1]) \cong \mathbb{D}_Y(i_y^*(Rf_*\underline{\mathbb{Q}}_X[1])) \cong (k^{\#f^{-1}(y)})^\vee[-1]$

$\Rightarrow P = Rf_*\underline{\mathbb{Q}}_X[1]$ is perverse.

$\mathrm{Hom}(P, i_{y*}V) = \mathrm{Hom}(i_y^*P, V) = \mathrm{Hom}(k^{\#f^{-1}(y)}[1], V) = 0$

& sim. $\mathrm{Hom}(i_{y*}V, P) = \mathrm{Hom}(V, (k^{\#f^{-1}(y)})^\vee[-1]) = 0$ (wrong degs! No EXT's between v.s.'s)

$\Rightarrow Rf_*\underline{\mathbb{Q}}_X[1] = j_{!*}\mathcal{L}[1]$.

Semisimplicity: $\pi_1(U,v) \longrightarrow S_d \subseteq GL(k^d)$ is a perm^n rep.

Let $G \subset S_d$ be image. By Maschke's thm, rep^n is SS. //

A more complicated example:

$f: E = \{y^2 = x(x-1)(x-\lambda)\} \xrightarrow{\lambda} \mathbb{A}^1$

By Hard Lefschetz: $Rf_*\underline{\mathbb{Q}}_E = \bigoplus_i R^i f_*\underline{\mathbb{Q}}_E[-i]$

- easy to check $R^0 f_*\mathbb{Q}_E = R^2 f_*\underline{\mathbb{Q}}_E = \underline{\mathbb{Q}}_E$
- We computed $R^1 f_*\underline{\mathbb{Q}}_E|_{\mathbb{A}^1 - \{0,1\}}$ has monodromy $\begin{pmatrix}1&0\\2&1\end{pmatrix}$ & $\begin{pmatrix}1&2\\0&1\end{pmatrix}$ $\Rightarrow$ Simple local system (no simultaneous fixed pts under these)
- We also checked $R^1 f_*\underline{\mathbb{Q}}_E = IC(R^1 f_*\underline{\mathbb{Q}}_E|_{\mathbb{A}^1-\{0,1\}})[-1]$.
  ↳ involved checking $H^1(E_0) = H^1(E_2)^{M_0}$
  $H^1(E_1) = H^1(E_2)^{M_1}$

SAVE FOR LATER (ERROR IN DECOMPOSING INTO CON. SHEAVES!)

Ex: $f: Bl_0(\mathbb{A}^2) \longrightarrow \mathbb{A}^2$: proper base change $\Rightarrow (R^j f_*\underline{\mathbb{Q}}_X)_0 \cong H^j(\mathbb{P}^1, \mathbb{Q})$

consid fibers $\Rightarrow R^0 f_*\underline{\mathbb{Q}}_X = \underline{\mathbb{Q}}_{\mathbb{A}^2}$. $R^2 f_*\underline{\mathbb{Q}}_X = i_*\mathbb{Q}$. OVER→

$= \begin{cases} \mathbb{Q}, & j = 0, 2 \\ 0, & \text{else.} \end{cases}$

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So by Hard Lefschetz + splitting:

$$Rf_*\underline{\mathbb{Q}}_X \cong \underline{\mathbb{Q}}_{\mathbb{A}^2} \oplus i_*\mathbb{Q}[-2]$$

($\underline{\mathbb{Q}}_{\mathbb{A}^2}[1]$ is simple, perv.)

($i_*\mathbb{Q}$ is simple, perverse.)

} note this is splitting into perverse coh., not just usual cohs, (since map is not smooth, this is important!)

All these examples point towards the D.T.:

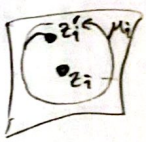
Decomposition Thm: $f: X \to Y$ proper; X: smooth. Then $Rf_*\underline{\mathbb{Q}}_X$ is a direct sum of shifts of simple perverse sheaves.

⇝ Can often detect behavior on sing. pts. w/info. on sm. pts.

Cor/Thm: (Local Invariant Cycles). $f: X \to C$ proper, X: smooth, C: sm. curve. $z_1, \dots, z_m$ = sing. pts. of f.

μ_i = loop around z_i (& only z_i) based @ z_i' close to z_i

Then ✡ $H^i(X_{z_i}) \longrightarrow H^i(X_{z_i'})^{\mu_i}$ is surjective.



Pf: $j_{!*}\mathcal{L} = \underbrace{j_*\mathcal{L}}_{\text{underived}}$ $\qquad (j_*\mathcal{L})_{z_i} = \ker(\mu_i - 1) = H^i(X_{z_i'})^{\mu_i}$

$$Rf_*\underline{\mathbb{Q}}_X \overset{\text{D.T.}}{\cong} \bigoplus_q j_*(\mathcal{L}^q)[-q] \oplus \bigoplus_q i_* W^q[-q]$$

($\mathcal{L}^q = {}^pR^qf_*\underline{\mathbb{Q}}_X|_U$)

✡ becomes projection away from W^q terms.

- Decomp^n Thm. is hard: will require Hodge theory/char. p to tackle. (We will use De Cataldo–Migliorini: simultaneously prove D.T. & H.L. via induction.)
- Doesn't generally control supports, ie. what supps. Z can appear as $IC(Z, \mathcal{L})$ for a simple factor of $Rf_*\underline{\mathbb{Q}}_X$.

⇝ **Rmk:** This is a first pass @ "nearby & vanishing cycles"

$$H^i(X_{z_i'})^{\mu_i} = j_{!*}\mathcal{L}(D^*)$$

(D = disk @ z_i)